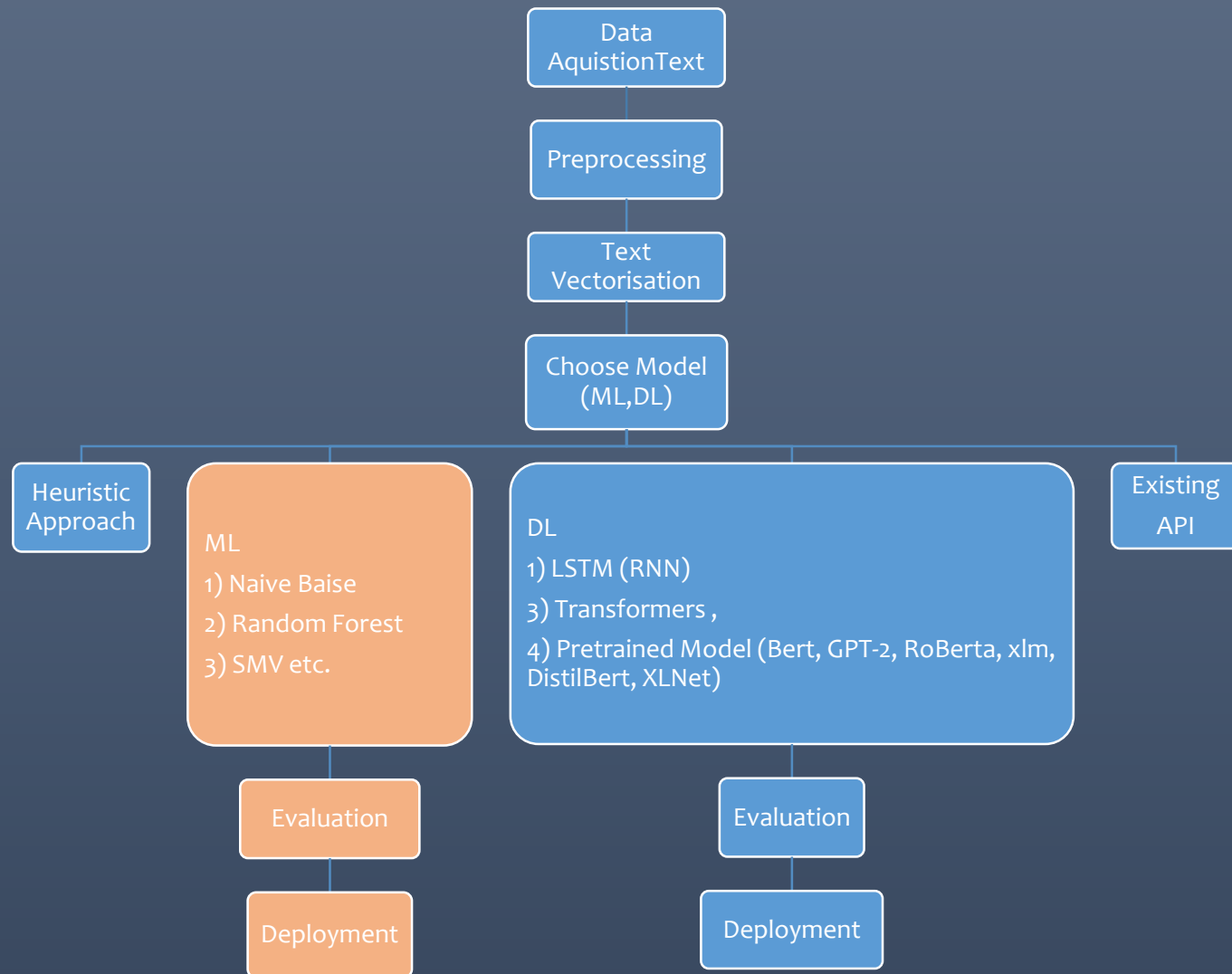




- 1) Data Acquisition
- 2) Data Augmentation
- 3) Text Preparation
  - a. Text Cleaning:
  - b. Pre-processing
- 4) Text Representation (Feature Engineering/Text Vectorisation/ Word Embedding's):
- 5) Model Building
- 6) Deployment

## Text Preparation:

There are total three ways of text preparation

- a) **Text Cleaning:**
- b) **Pre-processing**

### Text Cleaning:

Removing of unnecessary or irrelevant information that could hinder analysis or model performance.

- Lower casing
- Html Tag Cleaning
- Remove links
- Remove punctuation
- Chat Word treatment
- Spelling checker
- Stop-word removal
- Emoji Cleaning
  - REMOVE Emoji
  - Replace with meaning of Emoji

### Pre-processing

Pre-processing in NLP involves several techniques that transform raw text into a structured format suitable for analysis or modelling.

These techniques include:

- **Tokenization**
- **Stemming:**
- **Lemmatization:**
- **Part-of-Speech (POS) Tagging:**

**Tokenization:** Tokenization breaks text into smaller units, such as sentences (sentence tokenization) or words (word tokenization), which are essential for subsequent analysis.



## Stemming:

Stemming reduces words to their root form, even if the root itself may not be a valid word, helping to capture variations of words. Stemming is faster as compared to Lemmatization.

## Lemmatization:

Lemmatization is similar to stemming but ensures words are transformed to their canonical forms based on context and meaning, resulting in valid words. It is little slow as compared to Stemming

### Stemming vs Lemmatization



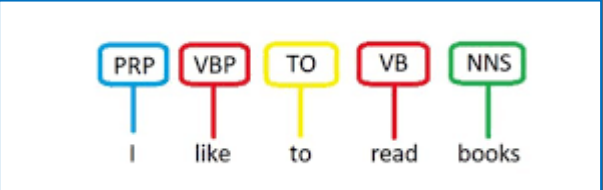
| Original Word | Stemmed Form | Lemmatized Form |
|---------------|--------------|-----------------|
| Playing       | Play         | play            |
| Plays         |              |                 |
| Played        |              |                 |
| Cats          | cat          | cat             |
| Happier       | happi        | happy           |
| Happiest      |              |                 |
| Easily        | easili       | easily          |
| Houses        | hous         | house           |
| Crying        | cri          | cry             |
| Cried         |              |                 |
| Strangely     | strang       | strangely       |
| Becoming      | becom        | become          |

### Stemming vs Lemmatization



**Part-of-Speech (POS) Tagging:** POS tagging assigns grammatical categories (like noun, verb, adjective) to words in a sentence, crucial for syntactic analysis and understanding sentence structure.

**Sentence:** "She walks quickly to the store to buy groceries."



| Word      | POS Tag |
|-----------|---------|
| She       | PRP     |
| walks     | VBZ     |
| quickly   | RB      |
| To        | TO      |
| the       | DT      |
| store     | NN      |
| to        | TO      |
| buy       | VB      |
| groceries | NNS     |
| .         | .       |

- PRP : Personal pronoun (She)
- VBZ : Verb, 3rd person singular present (walks)
- RB : Adverb (quickly)
- TO : to (preposition)
- DT : Determiner (the)
- NN : Noun, singular or mass (store)
- VB : Verb, base form (buy)
- NNS : Noun, plural (groceries)
- . : Punctuation mark (period)

## Feature Engineering || Text Representation || Text Vectorisation || Word Embedding

Feature engineering in natural language processing (NLP) involves transforming raw text data into numerical features that can be effectively used by machine learning models.

Here are some common feature engineering techniques used in NLP:

### Types

#### 1) Frequency-based

- **OHE**
- **Bow**
- **N-grams**
- **Tfidf,**
- GloVe
- Matrix
- fastText

#### 2) Prediction based: this is done based on prediction of the algorithms prediction

- word2vec
  - A) Continuous Bag of Words (CBOW):
  - B) Skip-gram:

One-Hot Encoding:

|    |                         |
|----|-------------------------|
|    | Example of comments     |
| D1 | People watch YouTuber   |
| D2 | YouTuber watch YouTuber |
| D3 | People write Comment    |
| D4 | YouTuber write Comment  |

|    | People | watch | YouTuber | write | Comment |
|----|--------|-------|----------|-------|---------|
| D1 | 1      | 0     | 0        | 0     | 0       |
|    | 0      | 1     | 0        | 0     | 0       |
|    | 0      | 0     | 1        | 0     | 0       |
|    | People | watch | YouTuber | write | Comment |
| D2 | 0      | 0     | 1        | 0     | 0       |
|    | 0      | 1     | 0        | 0     | 0       |
|    | People | watch | YouTuber | write | Comment |
| D3 | 1      | 0     | 0        | 0     | 0       |
|    | 0      | 0     | 0        | 1     | 0       |
|    | 0      | 0     | 0        | 0     | 1       |
|    | People | watch | YouTuber | write | Comment |
| D4 | 0      | 0     | 1        | 0     | 0       |
|    | 0      | 0     | 0        | 1     | 0       |
|    | 0      | 0     | 0        | 0     | 1       |

|    |                                     |
|----|-------------------------------------|
| D1 | [[1,0,0,0,0][0,1,0,0,0][0,0,1,0,0]  |
| D2 | [[0,0,1,0,0][0,1,0,0,0]]            |
| D3 | [[1,0,0,0,0][0,0,0,1,0][0,0,0,0,1]] |
| D4 | [[0,0,1,0,0][0,0,0,1,0][0,0,0,0,1]] |

**Bag of Words (BoW):**

BoW represents text data as a matrix where rows correspond to documents and columns correspond to unique words (or n-grams) in the corpus. Each cell in the matrix represents the frequency of the word's occurrence in the corresponding document. BoW disregards the order of words in the document and only considers their frequencies.

|    |                         |
|----|-------------------------|
|    | Example of comments     |
| D1 | People watch YouTuber   |
| D2 | YouTuber watch YouTuber |
| D3 | People write Comment    |
| D4 | YouTuber write Comment  |

|    | People | watch | YouTuber | write | Comment |
|----|--------|-------|----------|-------|---------|
| D1 | 1      | 1     | 1        | 0     | 0       |
| D2 | 0      | 1     | 2        | 0     | 0       |
| D3 | 1      | 0     | 0        | 1     | 1       |
| D4 | 0      | 0     | 1        | 1     | 1       |

N-grams:

- 1) One-gram
- 2) Bi-gram
- 3) Tri-gram

N-grams are contiguous sequences of n words from a given text. They capture local context and dependencies between adjacent words. Both unigrams (single words) and higher-order n-grams can be used as features in NLP tasks.

|    | Example of comments     |
|----|-------------------------|
| D1 | People watch YouTuber   |
| D2 | YouTuber watch YouTuber |
| D3 | People write Comment    |
| D4 | YouTuber write Comment  |

|    | People | watch | YouTuber | write | Comment |
|----|--------|-------|----------|-------|---------|
| D1 | 1      | 1     | 1        | -     | -       |
| D2 | -      | 1     | 1        | -     | -       |
| D3 | 1      | -     | -        | 1     | 1       |
| D4 | 1      | -     | -        | 1     | 1       |

One-gram

|    | People<br>watch | watch<br>YouTuber | YouTuber<br>watch | People<br>write | write<br>Comment | YouTuber<br>write |
|----|-----------------|-------------------|-------------------|-----------------|------------------|-------------------|
| D1 | 1               | 1                 | 0                 | 0               | 0                | 0                 |
| D2 | 0               | 1                 | 1                 | 0               | 0                | 0                 |
| D3 | 0               | 0                 | 0                 | 1               | 1                | 0                 |
| D4 | 0               | 0                 | 0                 | 0               | 1                | 1                 |

Bi-gram

|    | People watch<br>YouTuber | YouTuber<br>watch<br>YouTuber | People write<br>Comment | YouTuber write<br>Comment |
|----|--------------------------|-------------------------------|-------------------------|---------------------------|
| D1 | 1                        | 0                             | 0                       | 0                         |
| D2 | 0                        | 1                             | 0                       | 0                         |
| D3 | 0                        | 0                             | 1                       | 0                         |
| D4 | 0                        | 0                             | 0                       | 1                         |

Tri-gram

## TF x IDF (Term Frequency-Inverse Document Frequency):

- TF-IDF calculates the importance of a word in a document relative to its importance in the entire corpus.

|    | Example of comments     |
|----|-------------------------|
| D1 | People watch YouTuber   |
| D2 | YouTuber watch YouTuber |
| D3 | People write Comment    |
| D4 | YouTuber write Comment  |

*TF = Inverse Document Frequency*

*IDF = Term Frequency*

$$TF(i,j) = \frac{\text{Term } i \text{ Frequency in document } j}{\text{Total words in Documents } j}$$

$$IDF(i) = \log\left(\frac{\text{Total Documents}}{\text{Documents with term } i}\right)$$

$$TF \text{ \& IDF Score} = TF * IDF$$

|    |    | People | watch | YouTuber | write | Comment |
|----|----|--------|-------|----------|-------|---------|
| TF | D1 | 0.33   | 0.33  | 0.33     | -     | -       |
|    | D2 | -      | 0.33  | 0.67     | -     | -       |
|    | D3 | 0.33   | -     | -        | 0.33  | 0.33    |
|    | D4 | 0.33   | -     | -        | 0.33  | 0.33    |

|     |              | People | watch | YouTuber | write | Comment |
|-----|--------------|--------|-------|----------|-------|---------|
| IDF | For all Docs | 1.12   | 1.12  | 1.30     | 1.12  | 1.12    |

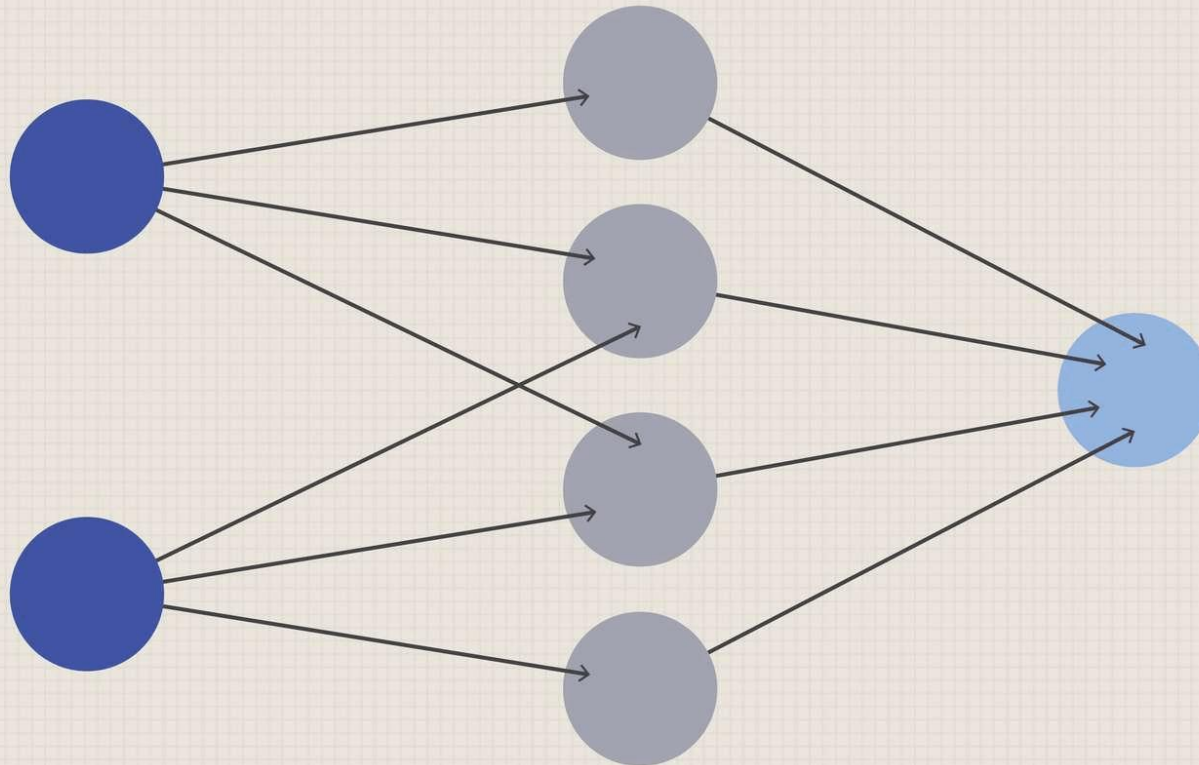
|          |    | People | watch | YouTuber | write | Comment |
|----------|----|--------|-------|----------|-------|---------|
| TF x IDF | D1 | 0.37   | 0.37  | 0.43     | -     | -       |
|          | D2 | -      | 0.37  | 0.87     | -     | -       |
|          | D3 | 0.37   | -     | -        | 0.37  | 0.37    |
|          | D4 | 0.37   | -     | -        | 0.37  | 0.37    |

## A Simple Neural Network

Input Layer

Hidden Layer

Output Layer



## Prediction based

- NNLM
- FastText
- Word2vec:

This is done based on prediction using neural network.

# Word2Vec

is a technique in natural language processing (NLP) that creates dense vector representations of words.

- These vectors capture semantic meanings by placing words with similar contexts closer together in a vector space. Unlike sparse representations,
- Word2Vec embedding's have fewer zero values because they are dense, meaning they store information efficiently.
- They typically have low dimensions (e.g., 100 to 300), making them computationally efficient while preserving meaningful relationships between words.

|        | King | Queen | Man | Women | Monkey |
|--------|------|-------|-----|-------|--------|
| Gender | 1    | 1     | 1   | 1     | 1      |
| Wealth | 1    | 1     | 0.3 | 0.3   | 0      |
| Power  | 1    | 0.7   | 0.2 | 0.2   | 0      |
| Weight | 0.8  | 0.4   | 0.6 | 0.5   | 0.3    |
| Speak  | 1    | 1     | 1   | 1     | 0      |

| King  | Man   | Women | King - Man + Women = Queen |       |       |       |
|-------|-------|-------|----------------------------|-------|-------|-------|
| 1     | 1     | 1     | 1                          | - 1   | + 1   | = 1   |
| 1     | 0.3   | 0.3   | 1                          | - 0.3 | + 0.3 | = 1   |
| 1     | 0.4   | 0.2   | 1                          | - 0.4 | + 0.2 | = 0.8 |
| 0.8   | 0.6   | 0.3   | 0.8                        | - 0.6 | + 0.3 | = 0.5 |
| 1     | 1     | 1     | 1                          | - 1   | + 1   | = 1   |
|       |       |       |                            |       |       |       |
| Queen | Women | Man   | Queen - Women + Man = King |       |       |       |
| 1     | 1     | 1     | 1                          | - 1   | + 1   | = 1   |
| 1     | 0.3   | 0.3   | 1                          | - 0.3 | + 0.3 | = 1   |
| 0.7   | 0.2   | 0.2   | 0.7                        | - 0.2 | + 0.2 | = 0.7 |
| 0.4   | 0.5   | 0.6   | 0.4                        | - 0.5 | + 0.6 | = 0.5 |
| 1     | 1     | 1     | 1                          | - 1   | + 1   | = 1   |

## Type of Word2Vec

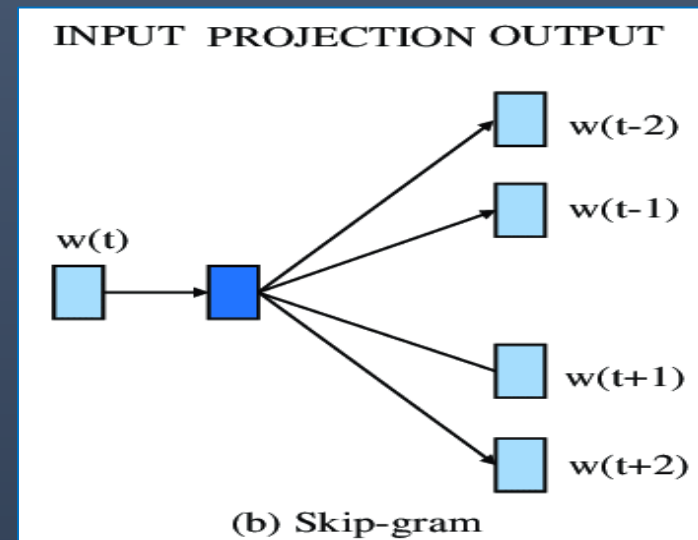
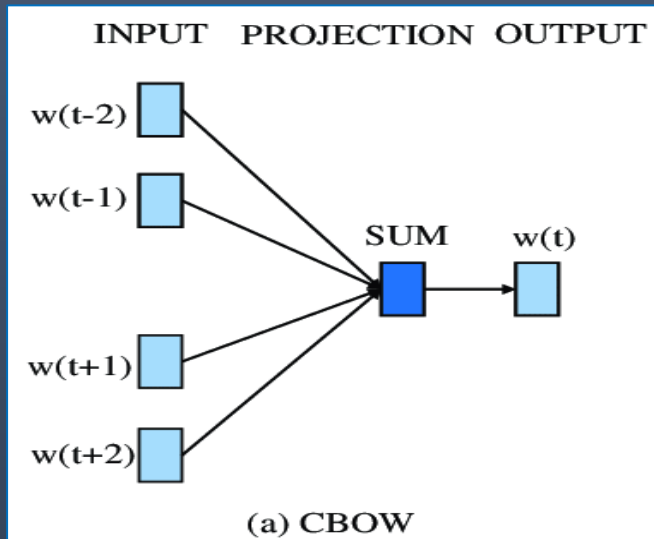
- 1) Continuous Bag of Words (CBOW):
- 2) Skip-gram:

**Continuous Bag of Words (CBOW):** Predicts a missing/target word using the surrounding words.

**Skip-gram:** Skip-Gram predicts the surrounding/context words from a given target word.

Slide 1 Watch \_\_\_\_\_ for Data Science

Slide 1 \_\_\_\_\_ YouTube \_\_\_\_\_ Data Science



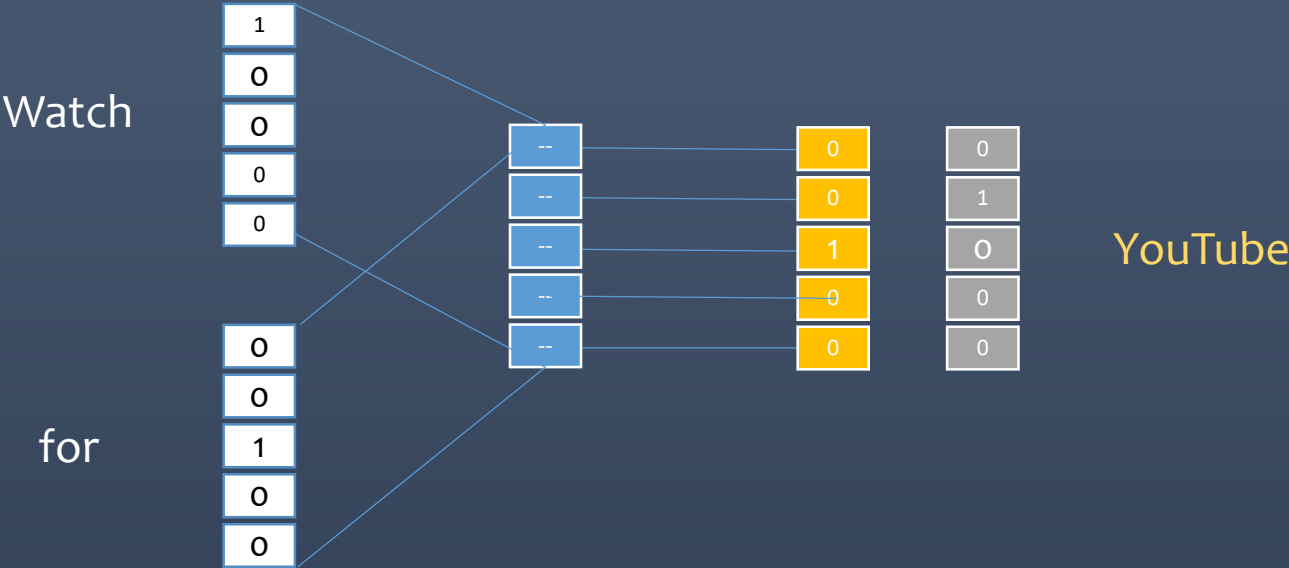
D1 Watch YouTube for Data Science

|         | Watch | YouTube | for | Data | Science |
|---------|-------|---------|-----|------|---------|
| Watch   | 1     | 0       | 0   | 0    | 0       |
| YouTube | 0     | 1       | 0   | 0    | 0       |
| for     | 0     | 0       | 1   | 0    | 0       |
| Data    | 0     | 0       | 0   | 1    | 0       |
| Science | 0     | 0       | 0   | 0    | 1       |

|        | King | Queen | Man | Women | Monkey |
|--------|------|-------|-----|-------|--------|
| Gender | 1    | 1     | 1   | 1     | 1      |
| Wealth | 1    | 1     | 0.3 | 0.3   | 0      |
| Power  | 1    | 0.7   | 0.2 | 0.2   | 0      |
| Weight | 0.8  | 0.4   | 0.6 | 0.5   | 0.3    |
| Speak  | 1    | 1     | 1   | 1     | 0      |

Slide 1 Watch YouTube for Data Science  
Slide 2 Watch YouTube for Data Science  
Slide 3 Watch YouTube for Data Science

|         | X1      | X2      | Y       |
|---------|---------|---------|---------|
| Slide 1 | Watch   | for     | YouTube |
| Slide 2 | YouTube | Data    | for     |
| Slide 3 | for     | Science | Data    |



D1 Watch YouTube for Data Science

|         | Watch | YouTube | for | Data | Science |
|---------|-------|---------|-----|------|---------|
| Watch   | 1     | 0       | 0   | 0    | 0       |
| YouTube | 0     | 1       | 0   | 0    | 0       |
| for     | 0     | 0       | 1   | 0    | 0       |
| Data    | 0     | 0       | 0   | 1    | 0       |
| Science | 0     | 0       | 0   | 0    | 1       |

|        | King | Queen | Man | Women | Monkey |
|--------|------|-------|-----|-------|--------|
| Gender | 1    | 1     | 1   | 1     | 1      |
| Wealth | 1    | 1     | 0.3 | 0.3   | 0      |
| Power  | 1    | 0.7   | 0.2 | 0.2   | 0      |
| Weight | 0.8  | 0.4   | 0.6 | 0.5   | 0.3    |
| Speak  | 1    | 1     | 1   | 1     | 0      |

Slide 1 Watch YouTube for Data Science  
Slide 2 Watch YouTube for Data Science  
Slide 3 Watch YouTube for Data Science

|         | x       | Y1      | Y2      |
|---------|---------|---------|---------|
| Slide 1 | YouTube | Watch   | For     |
| Slide 2 | for     | YouTube | Data    |
| Slide 3 | Data    | for     | Science |

